

Exp No: 24

MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

PROGRAM:

```
#include <systemc>

using namespace sc_core;
using namespace std;

SC_MODULE(ClockExample) {
    sc_in<bool> clk;

    void process() {
        while (true) {
            wait();
            cout << "Time: " << sc_time_stamp()
                << " - Clock Triggered" << endl;
        }
    }

    SC_CTOR(ClockExample) {
        SC_THREAD(process);
        sensitive << clk.pos();
    }
};

int sc_main(int argc, char* argv[]) {
    sc_clock clock("clock", 1, SC_NS);

    ClockExample module("module");
```

```
module.clk(clock);
```

```
sc_start(5, SC_NS);
```

```
return 0;
```

```
}
```

OUTPUT

The screenshot displays the EDA Playground web interface. The browser address bar shows the URL <https://www.edaplayground.com/x/5Vg4>. The interface includes a sidebar with navigation options like 'Languages & Libraries', 'Tools & Simulators', and 'Examples'. The main workspace is divided into two code editors: 'testbench.cpp' and 'design.cpp'. The 'testbench.cpp' editor contains the following code:

```
13         << " - Clock Triggered" << endl;
14     }
15
16     SC_CTOR(ClockExample) {
17         SC_THREAD(process);
18         sensitive << clk.pos();
19     }
20 };
21
22 int sc_main(int argc, char* argv[]) {
23     sc_clock clock("clock", 1, SC_NS);
24
25     ClockExample module("module");
26     module.clk(clock);
27
28     sc_start(5, SC_NS);
29
30     return 0;
31 }
```

The 'design.cpp' editor contains the following code:

```
1 // Code your design here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
```

Below the code editors is a 'Log' section showing the simulation output:

```
[2026-08-25 03:57:49 UTC] g++ -o sim -isystemc *.cpp && echo "compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
Done
```

The bottom of the image shows a Windows taskbar with various application icons and system status indicators, including the date and time (09:24 25-08-2026).